

## NEW ORLEANS NAS, LA, USA

WMO: 722316

Lat: 29.817N

Lon: 90.017W

Elev: 1

StdP: 101.32

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12958

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -0.8       | 1.2 | -8.6                          | 1.8 | 4.1  | -6.0 | 2.3 | 5.2  | 9.4                   | 20.7 | 8.7 | 19.2 | 1.9                   | 10   | 0.367 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.5                    | 34.0            | 25.4 | 33.2 | 25.3 | 32.5 | 25.2 | 27.6                | 30.7 | 27.0 | 30.3 | 26.6 | 30.0 | 2.8                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.8                            | 22.4 | 28.8 | 26.2 | 21.6 | 28.5 | 25.7 | 20.9 | 28.2 | 87.0          | 30.7 | 84.8 | 30.3 | 82.6 | 29.9 | 31.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 8.5           | 7.5  | 6.6 | -3.6                       | 35.8 | 2.2                | 1.1 | -5.2                                               | 36.6 | -6.4       | 37.2 | -7.7       | 37.8 | -9.2       | 38.7 |    |    |
| (5)               |      |     | -4.8                       | 28.6 | 2.0                | 1.2 | -6.3                                               | 29.5 | -7.5       | 30.2 | -8.6       | 30.9 | -10.1      | 31.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 20.6        | 11.7  | 14.0  | 16.9  | 20.1  | 24.3  | 27.1  | 27.8  | 28.0  | 26.2  | 21.4  | 16.1  | 13.5  |  |  |
|                                                                        | DBStd       | 6.89        | 5.41  | 5.43  | 4.56  | 3.51  | 2.34  | 1.61  | 1.20  | 1.36  | 2.01  | 3.78  | 4.39  | 5.30  |  |  |
|                                                                        | HDD10.0     | 102         | 44    | 22    | 6     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 26    |  |  |
|                                                                        | HDD18.3     | 734         | 216   | 138   | 81    | 23    | 1     | 0     | 0     | 0     | 0     | 16    | 94    | 166   |  |  |
|                                                                        | CDD10.0     | 3976        | 96    | 134   | 219   | 304   | 442   | 512   | 551   | 558   | 486   | 353   | 187   | 134   |  |  |
|                                                                        | CDD18.3     | 1567        | 8     | 18    | 37    | 77    | 184   | 262   | 292   | 299   | 236   | 111   | 26    | 17    |  |  |
|                                                                        | CDH23.3     | 14213       | 17    | 51    | 152   | 468   | 1587  | 2686  | 2994  | 3084  | 2149  | 848   | 127   | 51    |  |  |
|                                                                        | CDH26.7     | 5149        | 0     | 1     | 6     | 60    | 478   | 1114  | 1226  | 1279  | 776   | 197   | 11    | 1     |  |  |
| (14) Wind                                                              | WSAvg       | 2.6         | 3.1   | 3.3   | 3.2   | 3.3   | 2.7   | 2.2   | 1.8   | 1.7   | 2.1   | 2.5   | 2.7   | 3.0   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1566        | 137   | 126   | 129   | 136   | 125   | 142   | 159   | 153   | 162   | 86    | 106   | 119   |  |  |
|                                                                        | PrecMax     | 2579        | 450   | 368   | 262   | 580   | 334   | 520   | 266   | 382   | 424   | 287   | 564   | 412   |  |  |
|                                                                        | PrecMin     | 933         | 23    | 9     | 17    | 4     | 4     | 9     | 62    | 42    | 30    | 0     | 7     | 39    |  |  |
|                                                                        | PrecStd     | 367         | 94    | 78    | 58    | 140   | 85    | 111   | 59    | 69    | 102   | 63    | 99    | 73    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 25.2  | 26.5  | 27.5  | 29.4  | 32.7  | 34.5  | 34.7  | 35.4  | 34.2  | 31.8  | 27.9  | 26.3  |  |  |
|                                                                        |             | MCWB        | 20.7  | 21.6  | 21.1  | 21.9  | 23.5  | 24.6  | 25.8  | 25.3  | 25.5  | 24.8  | 22.8  | 22.0  |  |  |
|                                                                        | 2%          | DB          | 23.5  | 24.9  | 26.3  | 28.3  | 31.3  | 33.3  | 33.7  | 34.1  | 32.6  | 30.1  | 26.3  | 24.9  |  |  |
|                                                                        |             | MCWB        | 19.8  | 20.9  | 20.6  | 22.0  | 23.6  | 24.9  | 25.6  | 25.6  | 25.0  | 23.3  | 21.6  | 21.1  |  |  |
|                                                                        | 5%          | DB          | 22.2  | 23.5  | 25.2  | 27.3  | 30.3  | 32.5  | 32.9  | 33.1  | 31.7  | 28.8  | 24.8  | 23.3  |  |  |
|                                                                        |             | MCWB        | 19.1  | 20.1  | 20.4  | 21.6  | 23.5  | 25.0  | 25.5  | 25.7  | 24.7  | 22.6  | 20.5  | 20.6  |  |  |
|                                                                        | 10%         | DB          | 20.4  | 22.2  | 23.8  | 26.1  | 29.3  | 31.7  | 32.0  | 32.2  | 30.8  | 27.6  | 23.3  | 21.8  |  |  |
|                                                                        |             | MCWB        | 18.0  | 19.4  | 19.8  | 20.8  | 23.2  | 24.9  | 25.5  | 25.6  | 24.5  | 22.0  | 19.7  | 19.6  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 22.2  | 23.6  | 23.4  | 24.4  | 26.3  | 27.8  | 28.4  | 28.2  | 27.7  | 26.3  | 24.5  | 23.5  |  |  |
|                                                                        |             | MCDB        | 23.7  | 25.1  | 25.5  | 26.8  | 29.9  | 31.0  | 30.9  | 31.3  | 30.3  | 29.5  | 26.4  | 25.0  |  |  |
|                                                                        | 2%          | WB          | 20.9  | 22.3  | 22.3  | 23.7  | 25.5  | 26.9  | 27.4  | 27.5  | 26.8  | 25.4  | 23.0  | 22.3  |  |  |
|                                                                        |             | MCDB        | 22.8  | 23.6  | 24.3  | 26.2  | 28.8  | 30.3  | 30.4  | 31.0  | 29.7  | 28.3  | 25.0  | 23.7  |  |  |
|                                                                        | 5%          | WB          | 19.8  | 21.2  | 21.5  | 22.9  | 24.9  | 26.5  | 26.8  | 26.9  | 26.2  | 24.5  | 21.8  | 21.1  |  |  |
|                                                                        |             | MCDB        | 21.5  | 22.8  | 23.8  | 25.6  | 28.2  | 30.0  | 30.1  | 30.5  | 29.3  | 27.1  | 23.7  | 22.7  |  |  |
|                                                                        | 10%         | WB          | 18.3  | 19.9  | 20.6  | 22.2  | 24.2  | 25.8  | 26.3  | 26.5  | 25.6  | 23.8  | 20.5  | 19.9  |  |  |
|                                                                        |             | MCDB        | 20.0  | 21.8  | 23.3  | 24.9  | 27.7  | 29.6  | 29.9  | 30.1  | 28.8  | 26.3  | 22.8  | 21.5  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 10.1  | 10.0  | 10.1  | 9.8   | 9.5   | 8.9   | 8.5   | 8.5   | 8.7   | 10.1  | 11.0  | 9.8   |  |  |
|                                                                        |             | MCDBR       | 10.3  | 9.4   | 9.7   | 9.5   | 10.0  | 9.9   | 9.6   | 9.6   | 9.7   | 10.2  | 10.8  | 9.8   |  |  |
|                                                                        | 5% WB       | MCWBR       | 7.3   | 6.1   | 5.4   | 4.9   | 4.1   | 3.5   | 3.5   | 3.4   | 3.8   | 5.0   | 7.0   | 6.8   |  |  |
|                                                                        |             | MCWBR       | 8.5   | 8.2   | 8.4   | 7.9   | 8.5   | 8.6   | 8.5   | 8.4   | 8.2   | 7.9   | 9.5   | 9.0   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.342 | 0.360 | 0.376 | 0.420 | 0.446 | 0.465 | 0.508 | 0.503 | 0.448 | 0.387 | 0.362 | 0.348 |  |  |
|                                                                        |             | taud        | 2.535 | 2.498 | 2.438 | 2.320 | 2.283 | 2.270 | 2.179 | 2.190 | 2.309 | 2.471 | 2.481 | 2.529 |  |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 894   | 906   | 910   | 877   | 851   | 829   | 793   | 794   | 828   | 864   | 861   | 867   |  |  |
|                                                                        |             | Edh at Noon | 88    | 99    | 111   | 130   | 135   | 136   | 149   | 146   | 125   | 100   | 92    | 85    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 3.02  | 3.58  | 4.83  | 5.83  | 6.39  | 6.08  | 5.68  | 5.36  | 5.11  | 4.54  | 3.46  | 2.70  |  |  |
|                                                                        | RadStd      | RadStd      | 0.25  | 0.46  | 0.41  | 0.49  | 0.33  | 0.48  | 0.42  | 0.45  | 0.39  | 0.47  | 0.31  | 0.24  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (25 neighbors) |  | +0.42 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +136    | +89     |

Nomenclature: See separate page